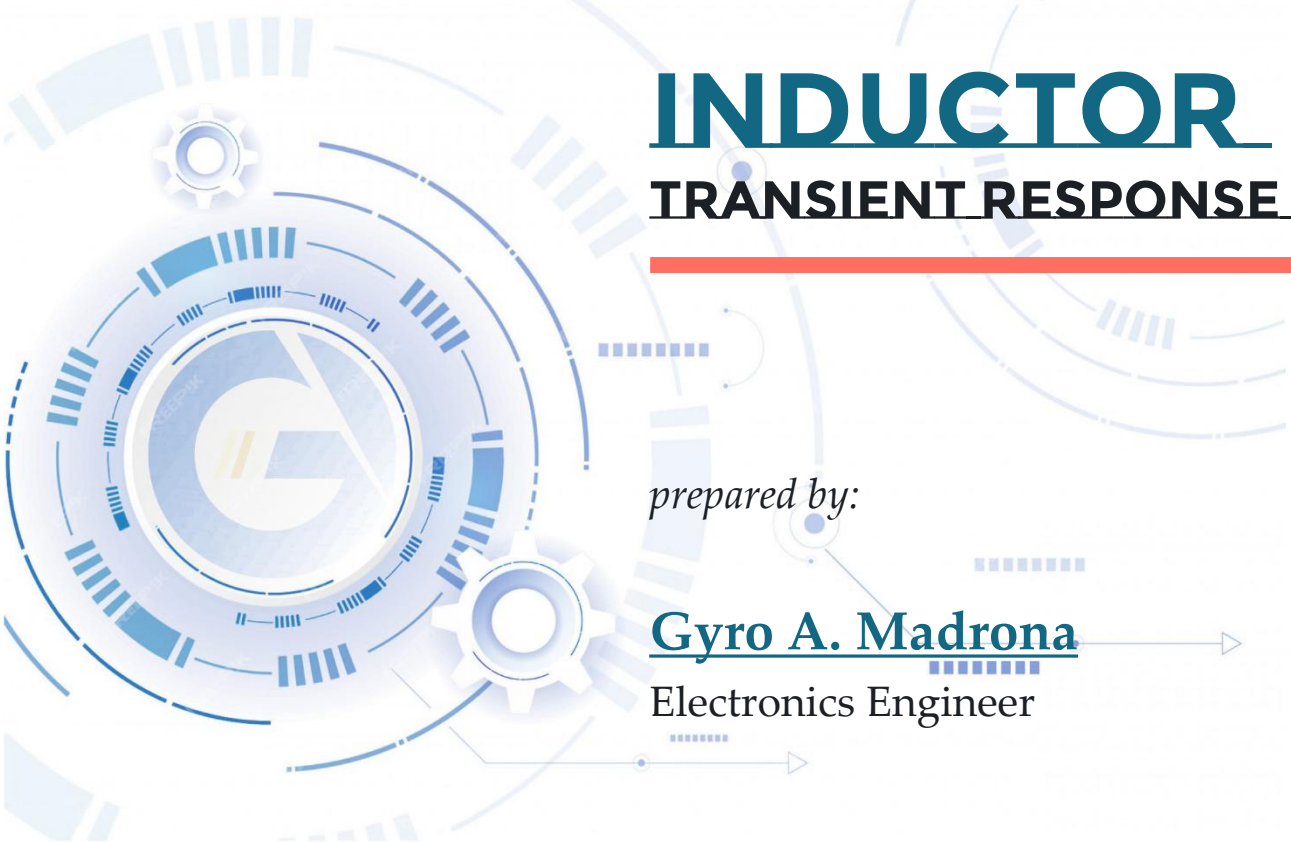




INDUCTOR

TRANSIENT RESPONSE

prepared by:

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TOPIC OUTLINE

RL Circuit

Energizing an Inductor

De-energizing an Inductor

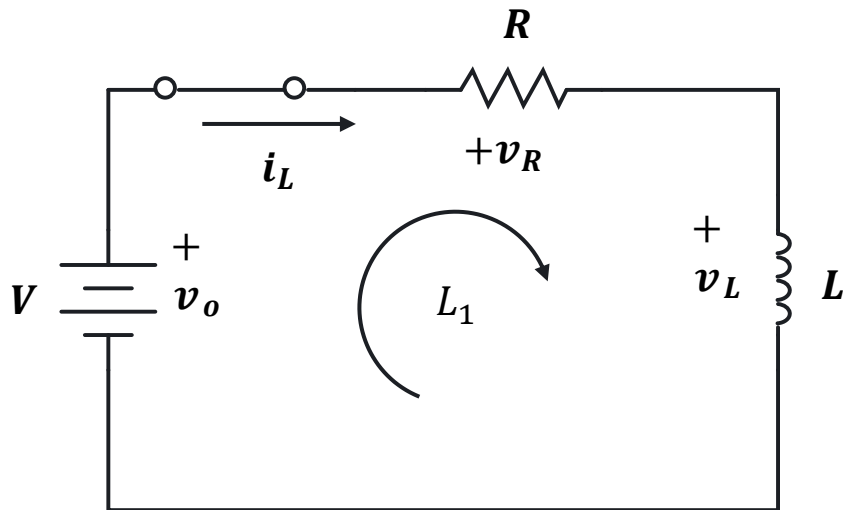
Transient Response



ENERGIZING AN INDUCTOR



RL CIRCUIT



KVL @ L_1

$$-v_o + v_R + v_L = 0$$

$$v_R + v_L = v_o$$

$$i_L R + v_L = v_o \quad ; v_L = L \frac{d}{dt} i_L$$

$$i_L R + L \frac{d}{dt} i_L = v_o$$

$$\frac{d}{dt} i_L + \frac{R}{L} i_L = \frac{v_o}{L}$$

... first-order ODE

$$i_L = \frac{v_o}{R} (1 - e^{-\frac{R}{L}t})$$



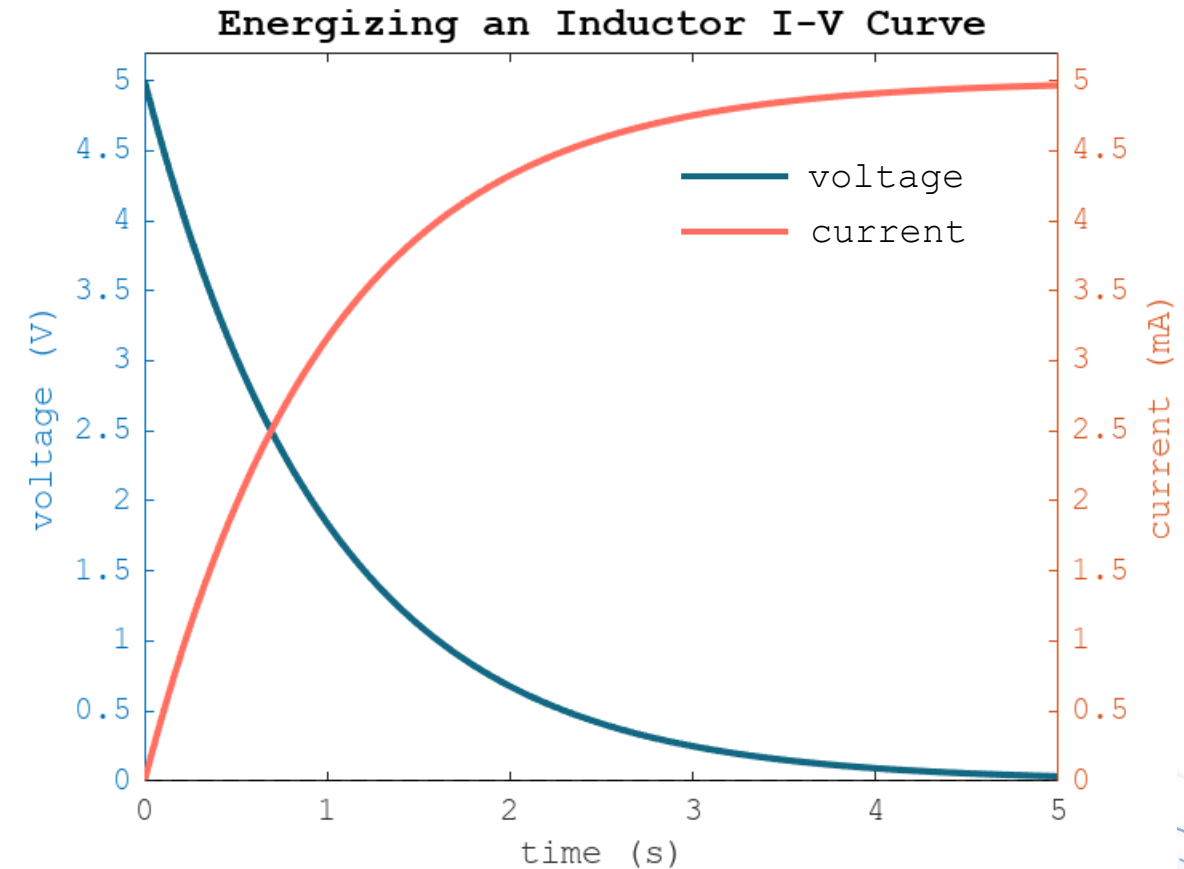
INDUCTOR CURRENT

Energizing equation

$$i_L(t) = \frac{v_o}{R} (1 - e^{-\frac{t}{\tau}})$$

where: $\tau = \frac{L}{R}$

The current through the inductor starts at zero and exponentially increases to $\frac{v_o}{R}$ amperes (maximum current).



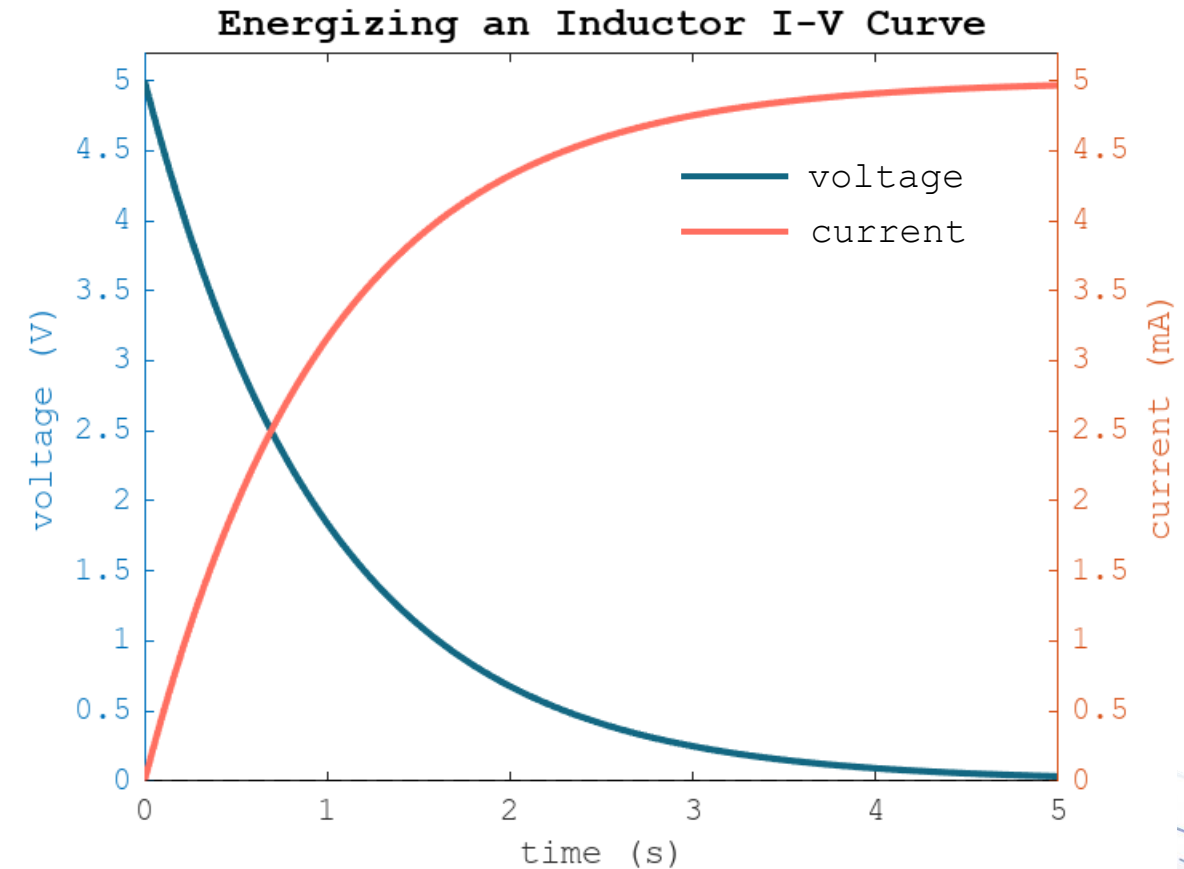
INDUCTOR VOLTAGE

Energizing equation

$$v_L(t) = v_o e^{-\frac{t}{\tau}}$$

$$\text{where: } \tau = \frac{L}{R}$$

The voltage across the inductor instantly jumps to its maximum value of v_o volts then decays exponentially to zero.



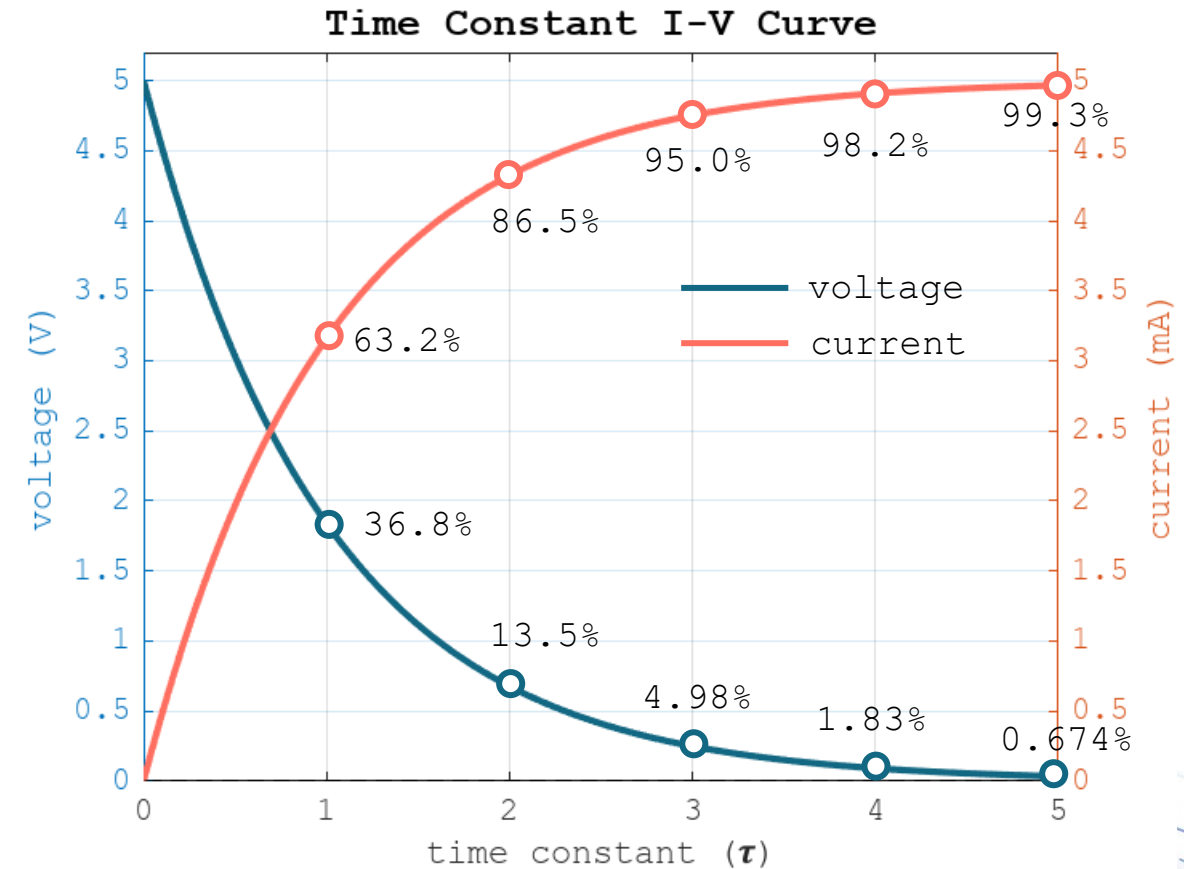
TIME CONSTANT

The time constant τ is a measure of how quickly an inductor energizes or de-energizes in an RL circuit.

Formula

$$\tau = \frac{L}{R}$$

unit: second



EXERCISE

A 50mH inductor is connected to a 12V DC power supply through a resistor of 500Ω . Determine the time it takes for the inductor to charge to 95% of its maximum current.

Solution



EXERCISE

A 50mH inductor is connected to a 12V DC power supply through a resistor of 500Ω . Determine the current through the inductor after $300\mu\text{s}$ of charging.

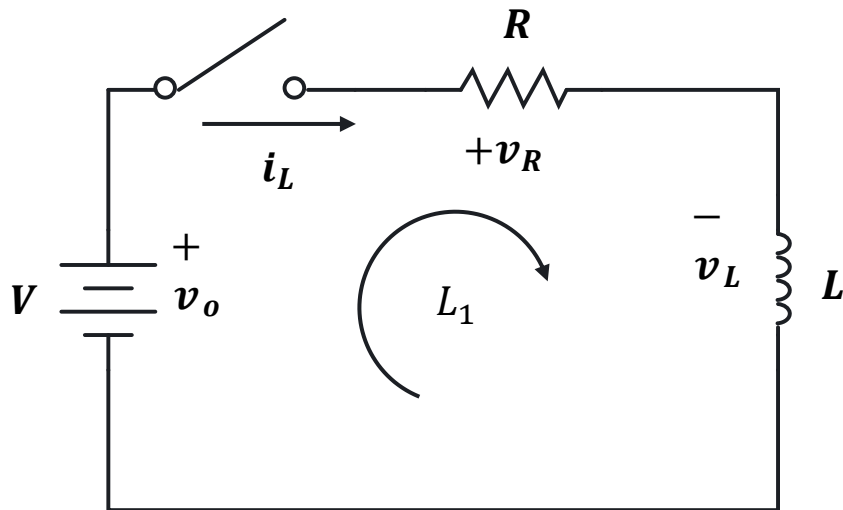
Solution



DE-ENERGIZING AN INDUCTOR



RL CIRCUIT



KVL @ L_1

$$v_R - v_L = 0$$

$$i_L R - v_L = 0 \quad ; v_L = L \frac{d}{dt} i_L$$

$$i_L R - L \frac{d}{dt} i_L = 0$$

$$\frac{d}{dt} i_L - \frac{R}{L} i_L = 0$$

... first-order ODE

$$i_L = \frac{v_o}{R} e^{-\frac{R}{L}t}$$



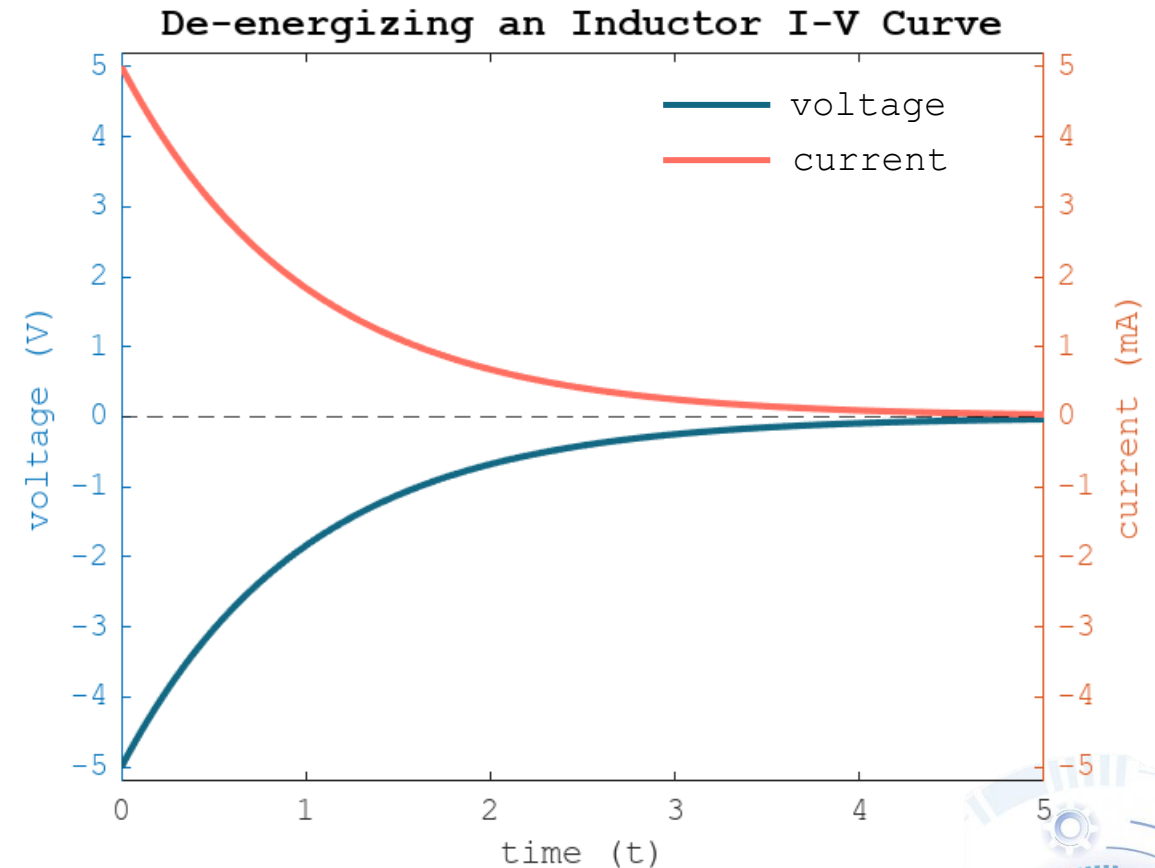
INDUCTOR CURRENT

De-energizing equation

$$i_L(t) = \frac{v_o}{R} e^{-\frac{t}{\tau}}$$

where: $\tau = \frac{L}{R}$

The current through the inductor starts at its maximum value $\frac{v_o}{R}$ amperes then decays exponentially to zero.



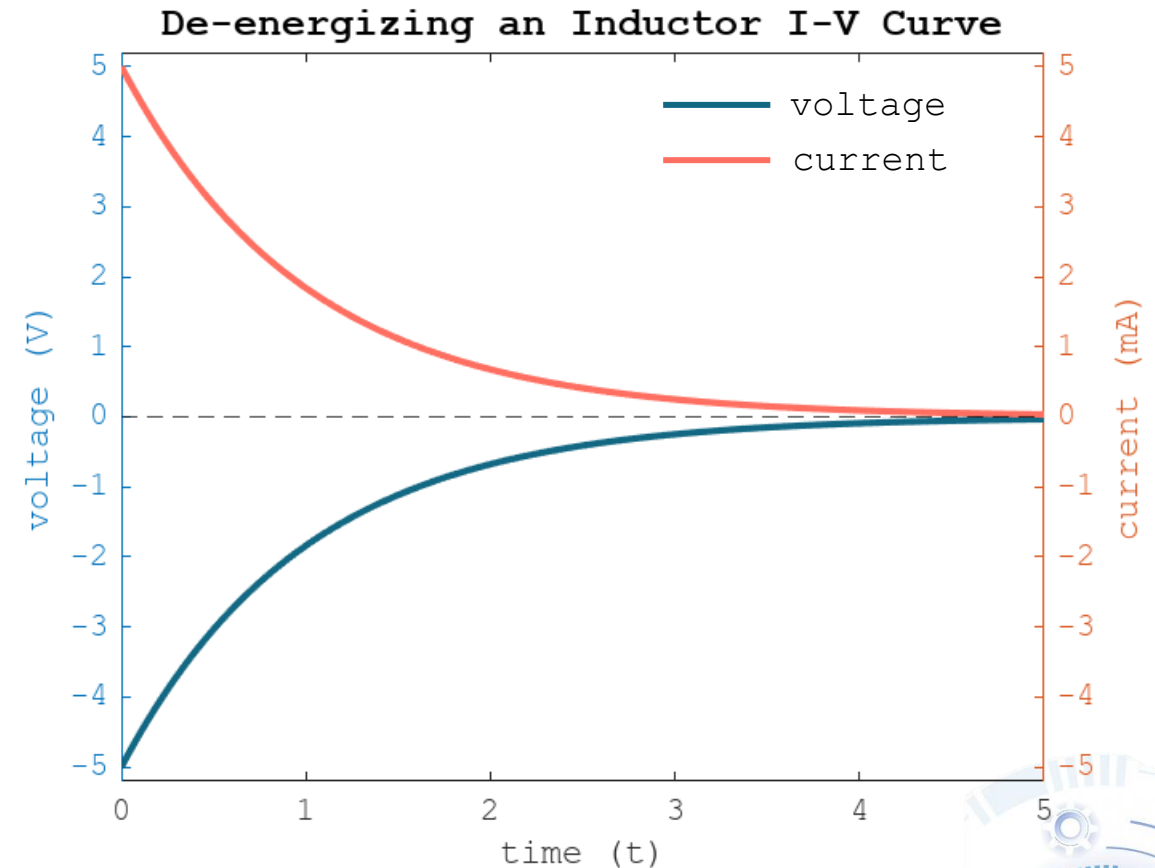
INDUCTOR VOLTAGE

De-energizing equation

$$v_L(t) = -v_o e^{-\frac{t}{\tau}}$$

where: $\tau = \frac{L}{R}$

The voltage across the inductor instantly jumps to its maximum value, but in opposite direction of $-v_o$ volts then decays exponentially to zero.



EXERCISE

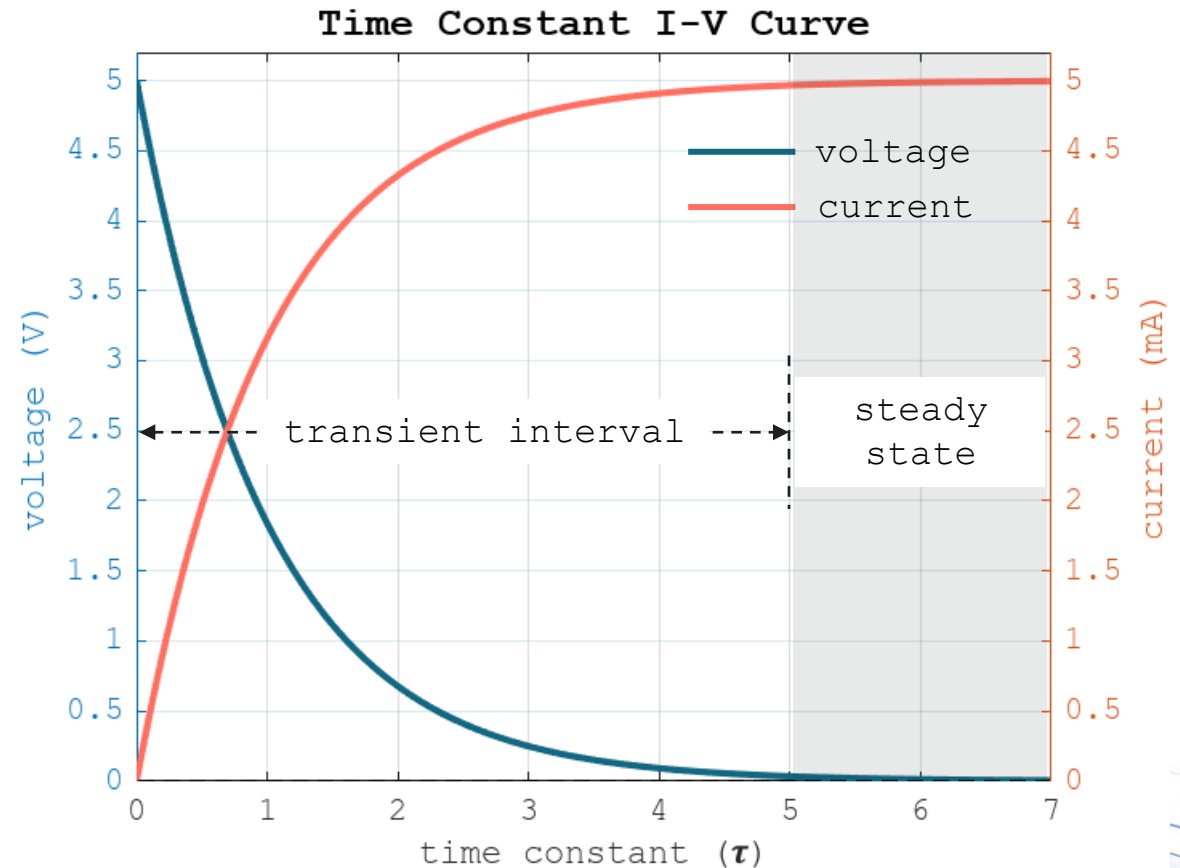
A 100mH inductor is initially charged to 6V . It is then disconnected from the power supply and discharged through a resistor of 500Ω . Determine the current through the inductor after 0.02s of discharging.

Solution



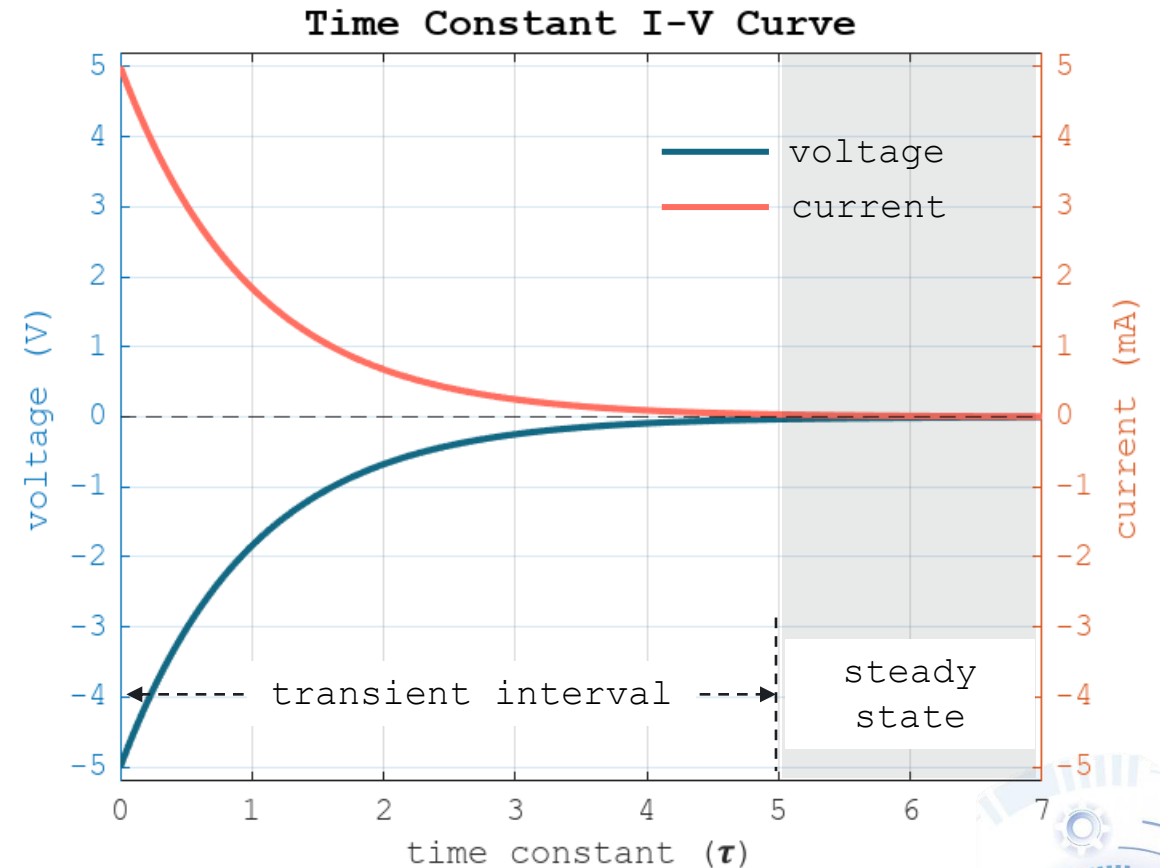
TRANSIENT RESPONSE

The transient response of an inductor describes the time-dependent changes in current through the inductor and the voltage across it. The transient phase is typically considered to last for approximately five time constants 5τ after which the system is assumed to have reached steady-state conditions.



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LABORATORY

